

REGULATORY DOSSIER SUBMISSION

Dossier ID: DOS-16B6AC14C4

Country: Tanzania

Submission Date: 2025-07-26

Product: Ampimed (ampicillin) | ATC: J01CA01 | Strength: 500 mg

Applicant: United Therapeutics | Manufacturer: Sahel Essential Drugs Company

Policy Label: standard_review | Risk Score: 0.5

AMR Stewardship: AWaRe=access | Unmet Need=routine | GLASS Trend=stable | Watch Similarity=low

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information

Labels: presence=present; length=length_ok; correctness=correct

This dossier constitutes a formal application for Marketing Authorization of Ampimed (ampicillin) submitted on 2025-07-26 to the national regulatory authority of Tanzania. The applicant, United Therapeutics, formally requests approval for the commercial distribution of this medicinal product. The proposed pharmaceutical form is capsule with a strength of 500 mg, classified under ATC code J01CA01. All required administrative declarations, legal attestations, letters of authorization, and regional application forms have been completed, signed by the responsible Qualified Person, and appended to Module 1. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Applicant authorization documents and legal attestations were reconciled against the named manufacturing parties. Method validation records and quality controls were reviewed for internal consistency. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The proposed prescribing information and patient information leaflet have been translated into local languages. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Pharmacokinetic parameters were

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence

Labels: presence=present; length=length_ok; correctness=correct

The primary commercial manufacturing site for the finished pharmaceutical product is Sahel Essential Drugs Company, located in Botswana. The facility was subject to a comprehensive GMP inspection by a recognized stringent regulatory authority. The latest inspection concluded on 2025-02-26, resulting in a formal compliance status of 'compliant'. The active GMP certificate number is GMP-156837, which remains valid until 2027-09-26. Site master files, a summary of recent inspection observations, and evidence of completed Corrective and Preventive Actions (CAPA) are included in the annex. Recent regulatory inspections yielded no critical observations, and all major findings have been effectively closed. Inspection history, CAPA closure records, and site responsibilities were reconciled across the manufacturing chain. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Method validation records and quality controls were reviewed for internal consistency. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The GMP evidence package includes certificate status, inspection scope, and manufacturing-site accountability records. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The

analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Batch manufacturing records and standard operating procedures (SOPs) are maintained and archived on-

Site Inspection History Summary:

Inspection Date	Authority	Scope	Outcome
2025-02-26	National Auth	FPP/General	compliant
2022-05-12	SRA Joint	API/Sterile	compliant

[1] m1_product_information - Product Information and Naming

Labels: presence=present; length=length_ok; correctness=correct

The proposed proprietary name for this medicinal product is Ampimed, containing the active pharmaceutical ingredient ampicillin. The draft Summary of Product Characteristics (SmPC), Patient Information Leaflet (PIL), and primary/secondary packaging labels are provided for review. These documents detail the approved therapeutic indications, posology, contraindications, special warnings, and precautions for use. A comprehensive risk management plan to minimize medication errors is also submitted. WHO AWaRe category: Access. Product is positioned for routine antibacterial use with standard stewardship monitoring and local resistance surveillance. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The applicant confirms alignment with applicable technical guidance and regional submis

[2] m2_quality_overall_summary - Quality Overall Summary

Labels: presence=present; length=length_ok; correctness=correct

This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for ampicillin. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. The applicant confirms alignment with applicable technical guidance and regional submission standards. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Method validation records and quality controls were reviewed for internal consistency. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. All referenced documents are available in the dossier annex and traceable by section identifier. Stability testing was conducted

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary

Labels: presence=present; length=length_ok; correctness=correct

The Clinical Overview synthesizes the efficacy and safety data supporting the use of Ampimed for the treatment of bacterial skin infection. The pivotal clinical development program comprised 2 well-controlled studies. Based on the integrated analysis, the reported clinical outcome category is 'endpoint_met'. The overall benefit-risk profile is considered highly favorable for the target patient population. Safety findings were consistent with the known pharmacological class effects, and appropriate risk minimization measures have been integrated into the proposed prescribing information. GLASS-aligned surveillance trend is stable, supporting continued first-line or second-line use with routine monitoring. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. No new safety signals were identified during the extended open-label follow-up period. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Stability testing was conducted in climatic zones III and IV.

[3] m3_api_quality - API Quality and Control Strategy

Labels: presence=present; length=length_ok; correctness=correct

The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The API package links specification limits to validated analytical methods and impurity-control decisions. All referenced documents are available in the dossier annex and traceable by section identifier. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The applicant confirms alignment with applicable technical guidance and regional submission standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Residual solvents and elemental impurities are controlled well below ICH Q3C and Q3D permissible daily exposures. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Method validation records and quality controls were reviewed for internal consistency. The synthetic route involves three well-characterized steps with defined starting materials and isolated intermediates. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Batch analysis, validation, and impurity management records support the proposed API quality framework. Critical material attributes and

control strategy elements were reviewed for consistency across the quality dossier. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covaria

[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls

Labels: presence=present; length=length_ok; correctness=correct

The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and reproducibility of the commercial manufacturing operations. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Method validation records and quality controls were reviewed for internal consistency. The applicant confirms alignment with applicable technical guidance and regional submission standards. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. All referenced documents are available in the dossier annex and traceable by section identifier. Method validation records and quality controls were reviewed for internal consistency. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The applicant confirms alignment with applicable technical guidance and regional submission standards. Statistical significance was set at an alpha level

[3] m3_stability - Stability and Shelf-Life Justification

Labels: presence=present; length=length_ok; correctness=correct

A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40°C/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. The applicant confirms alignment with applicable technical guidance and regional submission standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare

professionals. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. All referenced documents are available in the dossier annex and traceable by section identifier. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The manufacturing

[4] m4_nonclinical_summary - Nonclinical Study Summary

Labels: presence=present; length=length_ok; correctness=correct

The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The container closure system consists of PVC/PVDC-Aluminum blisters, com

[5] m5_trial_listing - Tabular Listing of Clinical Studies

Labels: presence=present; length=length_ok; correctness=correct

The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (bacterial skin infection). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. GLASS-aligned surveillance trend is stable, supporting continued first-line or second-line use with routine monitoring. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. All referenced documents are available in the dossier annex and traceable by section identifier. Method validation records and quality controls were reviewed for internal consistency. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Wher

Pivotal Clinical Trial Inventory:

Protocol ID	Phase	Enrollment	Status
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PROT-001	III	399	completed
PROT-002	III	732	completed

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports

Labels: presence=present; length=length_ok; correctness=correct

Detailed clinical study reports (CSRs) for the pivotal efficacy trials are provided. These randomized, double-blind, actively-controlled trials evaluated the primary and secondary endpoints in accordance with the pre-specified statistical analysis plan (SAP). The trials resulted in a formal outcome category of 'endpoint_met', demonstrating robust treatment effects in the intent-to-treat (ITT) and per-protocol populations. Subgroup analyses across age, gender, and baseline disease severity strata were consistent with the primary findings. Serious adverse events (SAEs) and discontinuations were systematically analyzed and adjudicated. GLASS-aligned surveillance trend is stable, supporting continued first-line or second-line use with routine monitoring. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The primary efficacy endpoint was met, demonstrating a statistically significant improvement over placebo ($p < 0.001$). Discontinuations due to treatment-emergent adverse events (TEAEs) were low and balanced across all study arms. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Method validation records and quality controls were reviewed for internal consistency. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The applicant confirms alignment with applicable technical guidance and regional submission standards. The pivotal study narratives align endpoint definitions, analysis populations, and outcome interpretation. Clinical study reports document protocol adherence, endpoint handling, and safety interpretation in the target population. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence

Labels: presence=present; length=length_ok; correctness=correct

Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of Cmax and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f_2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The analytical procedure was validated in a